

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

Student Name: K. Sirisha

Roll No.: A24126552260

Section: B

1. TITLE

Design a College Information Portal Webpage

2. AIM

To design a multi-section College Information Portal webpage using HTML5 that presents information about the college, its courses, students and registration facilities.

3. OBJECTIVE

- To understand the basic structure of an HTML5 document.
- To apply text formatting tags and organize content using headings, paragraphs and lists.
- To create hyperlinks, tables and insert images with alternate text.
- To design a student registration form using various form controls.
- To use HTML5 semantic elements for a well-structured layout.
- To embed audio and video using HTML5 media elements.
- To use HTML5 input types such as Email, Date, Number, Color and Range.

4. THEORY

HTML5 is the latest version of the Hyper Text Markup Language used to structure content on the web. It introduces semantic elements such as <header>, <nav>, <section>, <article> and <footer> that describe the meaning of the content they contain, making pages easier to read and maintain.

Text formatting tags such as , <i>, <u>, <sup> and <sub> are used to change how text is displayed. Lists are created using for ordered lists and for unordered lists. The <table> element, along with <tr>, <th> and <td>, is used to display tabular data such as student details.

HTML5 forms use elements like <input>, <select> and <textarea> to collect user data. New input types introduced in HTML5 — email, date, number, color and range — provide built-in validation and specialized controls in the browser. HTML5 also introduced native <audio> and <video> elements to embed media without external plugins.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare the HTML5 document type.
2. Add the <head> section with a page title.
3. Build the page structure using semantic tags — header, nav, main, section, article and footer.
4. Add headings and paragraphs describing the college, using text formatting tags.
5. Create one ordered list (courses) and one unordered list (facilities).
6. Add hyperlinks to the college website and to Google.
7. Insert the college logo and a relevant campus image with alternate text.
8. Create a table displaying student Roll No, Name, Branch and Email.

9. Design a registration form with text, email, password, radio, checkbox, dropdown and textarea fields.
10. Add HTML5 input types — Date, Number, Color and Range — inside the form.
11. Embed an audio player and a video player using the audio and video elements.
12. Save the file and open it in a web browser.
13. Verify that every section renders and functions as expected.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
  <header>
    <h1>College Information Portal</h1>
    <p>
      <b>Welcome to our college portal. </b>This webpage provides
      information about the college, students, courses, and
      registration facilities.
    </p>
  </header>
  <nav>
    <a href="#about">About College</a> |
    <a href="#courses">Courses</a> |
    <a href="#students">Students</a> |
    <a href="#registration">Registration</a> |
    <a href="#multimedia">Multimedia</a>
  </nav>
  <hr>
  <main>
    <section id="about">
      <h2>About Our College</h2>
      <article>
        <p>
          Our college provides <strong>quality education</strong>
          and encourages students to develop
          <em>technical and communication skills</em>. </p>
        <p>
          The college offers <u>modern learning facilities</u>
          and supports innovation and research.
        </p>
        <p>
          H<sub>2</sub>O is the chemical formula for water.
          The mathematical expression is x<sup>2</sup>.
        </p>
      </article>
      <h3>Useful Links</h3>
      <a href="https://www.anits.org" target="_blank">
        Visit College Website
      </a>
      <br><br>
      <a href="https://www.google.com" target="_blank">
        Open Google
      </a>
    </section>
    <hr>
    <section id="courses">
      <h2>Courses Offered</h2>
      <h3>Popular Courses</h3>
```

```
<ol>
<li>Computer Science and Engineering</li>
<li>Artificial Intelligence and Machine Learning</li>
<li>Electronics and Communication Engineering</li>
<li>Mechanical Engineering</li>
</ol>
<h3>College Facilities</h3>
<ul>
<li>Central Library</li>
<li>Computer Laboratories</li>
<li>Sports Ground</li>
<li>College Canteen</li>
</ul>
</section>
<hr>
<section id="images">
<h2>College Images</h2>


</section>
<hr>
<section id="students">
<h2>Student Details</h2>
<table border="1" cellpadding="10">
<caption><strong>Student Information</strong></caption>
<thead>
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Branch</th>
<th>Email</th>
</tr>
</thead>
<tbody>
<tr><td>101</td><td>Anusha</td><td>CSE</td><td>anusha@example.com</td></tr>
<tr><td>102</td><td>Rahul</td><td>AIML</td><td>rahul@example.com</td></tr>
<tr><td>103</td><td>Priya</td><td>ECE</td><td>priya@example.com</td></tr>
</tbody>
</table>
</section>
<hr>
<section id="registration">
<h2>Student Registration Form</h2>
<form id="regForm">
<label for="name">Student Name:</label>
<input type="text" id="name" name="name" placeholder="Enter your name" required>
<br><br>
<label for="email">Email:</label>
<input type="email" id="email" name="email" placeholder="Enter your email" required>
<br><br>
<label for="password">Password:</label>
<input type="password" id="password" name="password" required>
<br><br>
<p><strong>Gender:</strong></p>
<input type="radio" id="male" name="gender" value="Male">
<label for="male">Male</label>
<input type="radio" id="female" name="gender" value="Female">
<label for="female">Female</label>
<input type="radio" id="other" name="gender" value="Other">
<label for="other">Other</label>
<br><br>
<p><strong>Hobbies:</strong></p>
<input type="checkbox" id="reading" name="hobbies" value="Reading">
<label for="reading">Reading</label>
<input type="checkbox" id="sports" name="hobbies" value="Sports">
```

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<label for="sports">Sports</label>
<input type="checkbox" id="music" name="hobbies" value="Music">
<label for="music">Music</label>
<br><br>
<label for="branch">Select Branch:</label>
<select id="branch" name="branch">
<option value="">-- Select Branch --</option>
<option value="CSE">Computer Science and Engineering</option>
<option value="AIML">Artificial Intelligence and Machine Learning</option>
<option value="ECE">Electronics and Communication Engineering</option>
<option value="MECH">Mechanical Engineering</option>
</select>
<br><br>
<label for="dob">Date of Birth:</label>
<input type="date" id="dob" name="dob">
<br><br>
<label for="age">Age:</label>
<input type="number" id="age" name="age" min="16" max="100">
<br><br>
<label for="colour">Favourite Colour:</label>
<input type="color" id="colour" name="colour">
<br><br>
<label for="rating">Rate the College:</label>
<input type="range" id="rating" name="rating" min="1" max="10" value="5">
<br><br>
<label for="comments">Comments:</label>
<br>
<textarea id="comments" name="comments" rows="5" cols="40" placeholder="Enter your
comments"></textarea>
<br><br>
<input type="submit" value="Register">
<input type="reset" value="Clear Form">
</form>
</section>
<hr>
<section id="multimedia">
<h2>College Multimedia</h2>
<h3>College Audio</h3>
<audio controls>
<source src="college-audio.mp3" type="audio/mpeg">
Your browser does not support audio.
</audio>
<h3>College Video</h3>
<video width="500" controls>
<source src="college-video.mp4" type="video/mp4">
Your browser does not support video.
</video>
</section>
</main>
<hr>
<footer>
<p>&copy; 2026 College Information Portal. All rights reserved.</p>
</footer>
</body>
</html>

```

7. CODE EXPLANATION

Tag / Attribute	Explanation
<header>, <nav>, <main>, <section>, <article>, <footer>	HTML5 semantic elements used to structure the page into meaningful regions.
, , , <u>, <sup>, <sub>	Format text as bold, strongly emphasized, italic, underlined, superscript and subscript.
/	Display the courses list (ordered) and facilities list (unordered).
	Creates hyperlinks to the college website and Google.
	Displays the college logo and campus image; alt text shows if the image fails to load.
<table>, <caption>, <thead>, <tbody>, <tr>, <th>, <td>	Builds the student details table with Roll No, Name, Branch and Email.
<form>, <input>, <select>, <textarea>	Collects student registration data through various form controls.
type="radio" / "checkbox"	Lets the user select Gender (one option) and Hobbies (multiple options).
type="date" / "number" / "color" / "range"	HTML5 input types providing built-in pickers and validation in the browser.
<audio controls> / <video controls>	Embeds a playable audio element and a playable video element with default browser controls.

8. TEST CASES AND EXECUTION DATA

The webpage was opened in a browser and each section was verified against its expected behaviour. The output for each test case is shown right below it.

Test Case	TC-01
Action / Input	Load the page in browser
Expected Result	Header, nav bar, about section and lists display correctly
Actual Result	Displayed correctly
Status	Pass

TC-01 Output

College Information Portal

Welcome to our college portal. This webpage provides information about the college, students, courses, and registration facilities.

[About College](#) | [Courses](#) | [Students](#) | [Registration](#) | [Multimedia](#)

About Our College

Our college provides **quality education** and encourages students to develop *technical and communication skills*.

The college offers modern learning facilities and supports innovation and research.

H₂O is the chemical formula for water. The mathematical expression is x^2 .

Useful Links

[Visit College Website](#)

[Open Google](#)

Courses Offered

Popular Courses

1. Computer Science and Engineering
2. Artificial Intelligence and Machine Learning
3. Electronics and Communication Engineering
4. Mechanical Engineering

College Facilities

- Central Library
- Computer Laboratories
- Sports Ground
- College Canteen

Test Case	TC-02
Action / Input	Click "College Website" and "Google" links
Expected Result	Links open the respective pages in a new tab
Actual Result	Opened correctly
Status	Pass

TC-02 Output

Useful Links

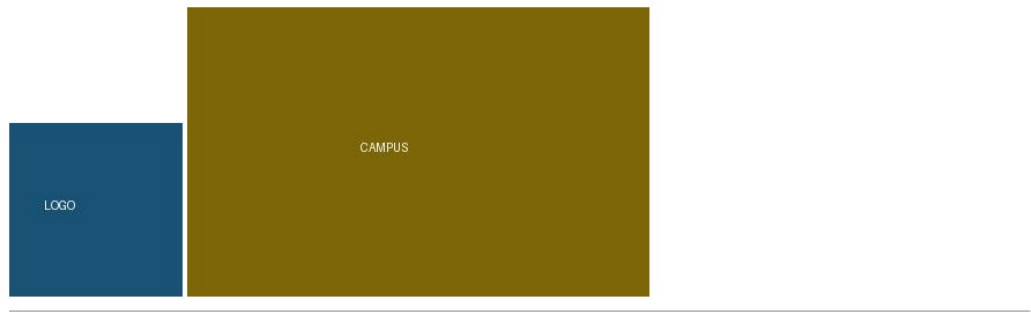
[Visit College Website](#)

[Open Google](#)

Test Case	TC-03
Action / Input	Check logo and campus image
Expected Result	Both images render with correct alternate text
Actual Result	Rendered correctly
Status	Pass

TC-03 Output

College Images



Test Case	TC-04
Action / Input	Check student details table
Expected Result	Roll No, Name, Branch and Email shown in a grid
Actual Result	Displayed correctly
Status	Pass

TC-04 Output

Student Details

Student Information			
Roll No	Name	Branch	Email
101	Anusha	CSE	anusha@example.com
102	Rahul	AIML	rahul@example.com
103	Priya	ECE	priya@example.com

Test Case	TC-05
Action / Input	Submit registration form with Name field empty
Expected Result	Browser should require the Name field
Actual Result	Browser blocked submission and asked to fill Name
Status	Pass

TC-05 Output

Student Name:

Email:

 Please fill out this field.

Password:

Test Case	TC-06
Action / Input	Select Gender and one or more Hobbies
Expected Result	Only one Gender selectable; multiple Hobbies selectable
Actual Result	Behaved correctly
Status	Pass

TC-06 Output

☐ Male ☒ Female ☐ Other

Hobbies:

☒ Reading ☐ Sports ☒ Music

Test Case	TC-07
Action / Input	Pick a Date, Number, Color and Range value
Expected Result	Native browser pickers open for each input type
Actual Result	Pickers opened as expected

Status	Pass
--------	------

TC-07 Output

Date of Birth: 05/14/2006

Age: 19

Favourite Colour:

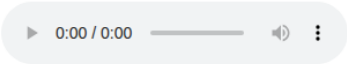
Rate the College:

Test Case	TC-08
Action / Input	Play the Audio and Video elements
Expected Result	Media controls (play/pause) are visible and functional
Actual Result	Controls displayed and functional
Status	Pass

TC-08 Output

College Multimedia

College Audio



College Video



9. OUTPUT

Output 1 — Header, About Section, Text Formatting, Lists, Hyperlinks, Facilities List, College Images, Student Table, Registration Form, Comments, Buttons, Media Elements & Footer

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Useful Links

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Courses Offered

Popular Courses

1. Computer Science and Engineering
2. Artificial Intelligence and Machine Learning
3. Electronics and Communication Engineering
4. Mechanical Engineering

College Facilities

- Central Library
- Computer Laboratories
- Sports Ground
- College Canteen

College Images

 College Logo  View of the college campus

Student Details

Student Information

Roll No	Name	Branch	Email
101	Anusha	CSE	anusha@example.com
102	Rahul	AIML	rahul@example.com
103	Priya	ECE	priya@example.com

Student Registration Form

Student Name:

Email:

Password:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Sports ☐ Music

Select Branch:

Date of Birth:

Age:

Favourite Colour:

Rate the College:

Comments:

College Multimedia

College Audio



College Video



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10. RESULT

Thus, a College Information Portal webpage was successfully designed using HTML5. Text formatting, lists, hyperlinks, tables, images, a registration form, semantic elements, HTML5 input types and embedded media were implemented, and the output was verified against the expected results for all test cases.